



User Query

Find a wireless noise-canceling headphone under \$300 with high rating.



Agent Trajectory τ



Search 1: headphone under \$300

Anchor A a

Rating: 4.1 $v_{a(c)}$
Price: \$199
Color: Black

Non-Anchor B

Rating: 4.8
Price: \$249
Color: Brown



Search 2: compare reviews

Anchor A

Rating: 4.1
Price: \$199
Color: Black

Non-Anchor C

Rating: 4.5
Price: \$279
Battery: 30 h

Final response: I would recommend product A is priced at \$199, has a rating of 4.8 and a 30-hour battery life

Anchor-Grounded Risk Estimation

Claim Hidden
State Extraction

Product A with
price \$199 and
rating <eos>

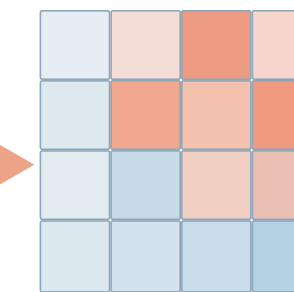


Frozen LLM Agent

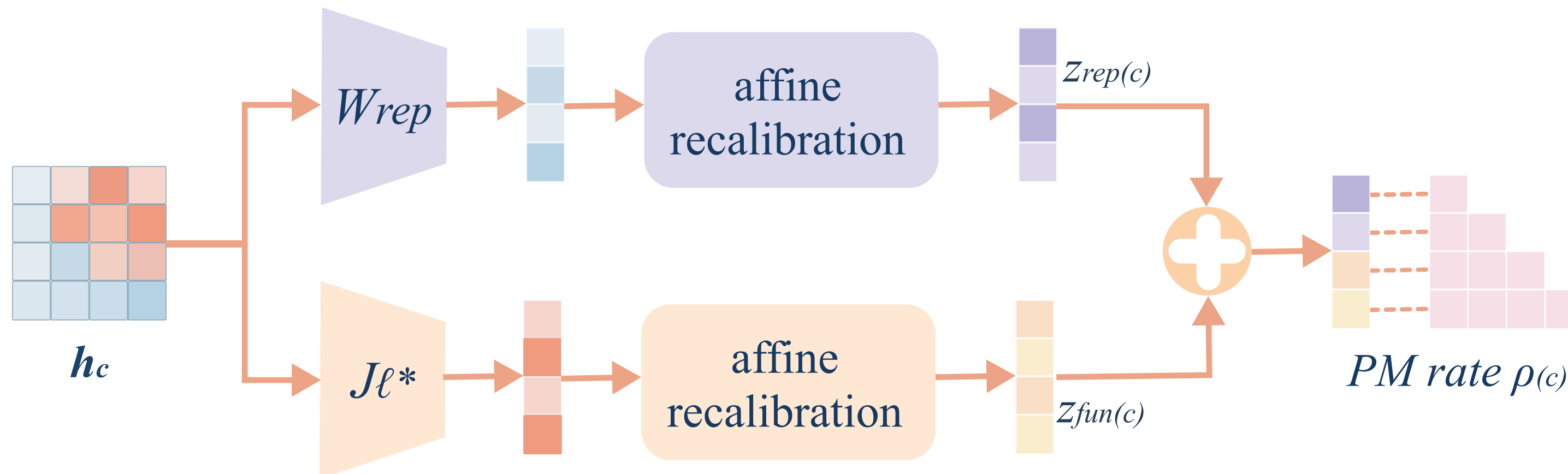
layer 1

layer ℓ^*

layer L



h_c



Fixed-Anchor Correction

Fixed Anchor:
I would
recommend
product A

KEEP:
 $\rho_{(c)} < \theta$
Product A is
priced at \$199

REWRITE:
 $\rho_{(c)} \geq \theta \wedge v_{a(c)}$
A has a rating of
4.8 $\rightarrow$ 4.1

REMOVE:
 $\rho_{(c)} \geq \theta \wedge \text{no } v_{a(c)}$
~~A has a 30-hour
battery life~~